

Oracle AI, Oracle Cloud Infrastructure (OCI), Technical Solutions

Simplifying Spatial Analytics with OCI Document Understanding and Select AI

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enterprises handle vast volumes of unstructured documents, where extracting meaningful —especially those tied to location data, zip codes with a combination of phone numbers, and numeric values—can be both time-intensive and error-prone.



Cloud Infrastructure (OCI) addresses this challenge by bringing together **Document AI**, **Select Spatial capabilities**, enabling developers to convert raw document data into actionable spatial intelligence.





Preface: This article demonstrates how to design and implement a spatial intelligence application on OCI by integrating the following:

OCI Document Understanding for intelligent document parsing

Oracle Autonomous Database for structured data persistence

Select AI (NL2SQL) for translating natural language into executable SQL queries

Oracle Spatial for advanced geospatial analysis

Together, these services form a cohesive pipeline that bridges unstructured document processing with location-aware analytics.

We'll go over how to construct a spatial data visualization application in this article using:

OCI Document Understanding

Oracle Autonomous Database (AI Database)

Select AI (Natural Language to SQL)

Spatial Analysis using pre-trained models

For visualization, either we can use Oracle APEX or Python Streamlit Map for a flexible, developer-

driven interface to render interactive maps and spatial insights.

What is OCI Document Understanding?

OCI Document Understanding is an AI service that enables developers to extract the following:

- Text
- Tables
- Key-value pairs
- Document classifications
- Optical Character Recognition (OCR)

It provides **pre-trained models** for common document processing tasks and supports customization for industry-specific needs. Refer to the [oracle document](#) for detailed information

Once a model is selected, documents (PDF/images) can be uploaded via APIs or CLI. and data from tables can be extracted either as **TEXT** or as **structured TABLE format (rows × columns)** with ease.

Solution Overview

For certain use cases where documents contain a large amount of numerical data—such as **pincodes, phone numbers, latitude, and longitude**—vectorizing digits may not yield optimal results.

In such scenarios, to achieve better accuracy and performance with LLMs, it is more effective to:

- Store the extracted data in **relational format**
- Use **NL2SQL (Select AI)** to query the data

With this approach, we build a pipeline that:

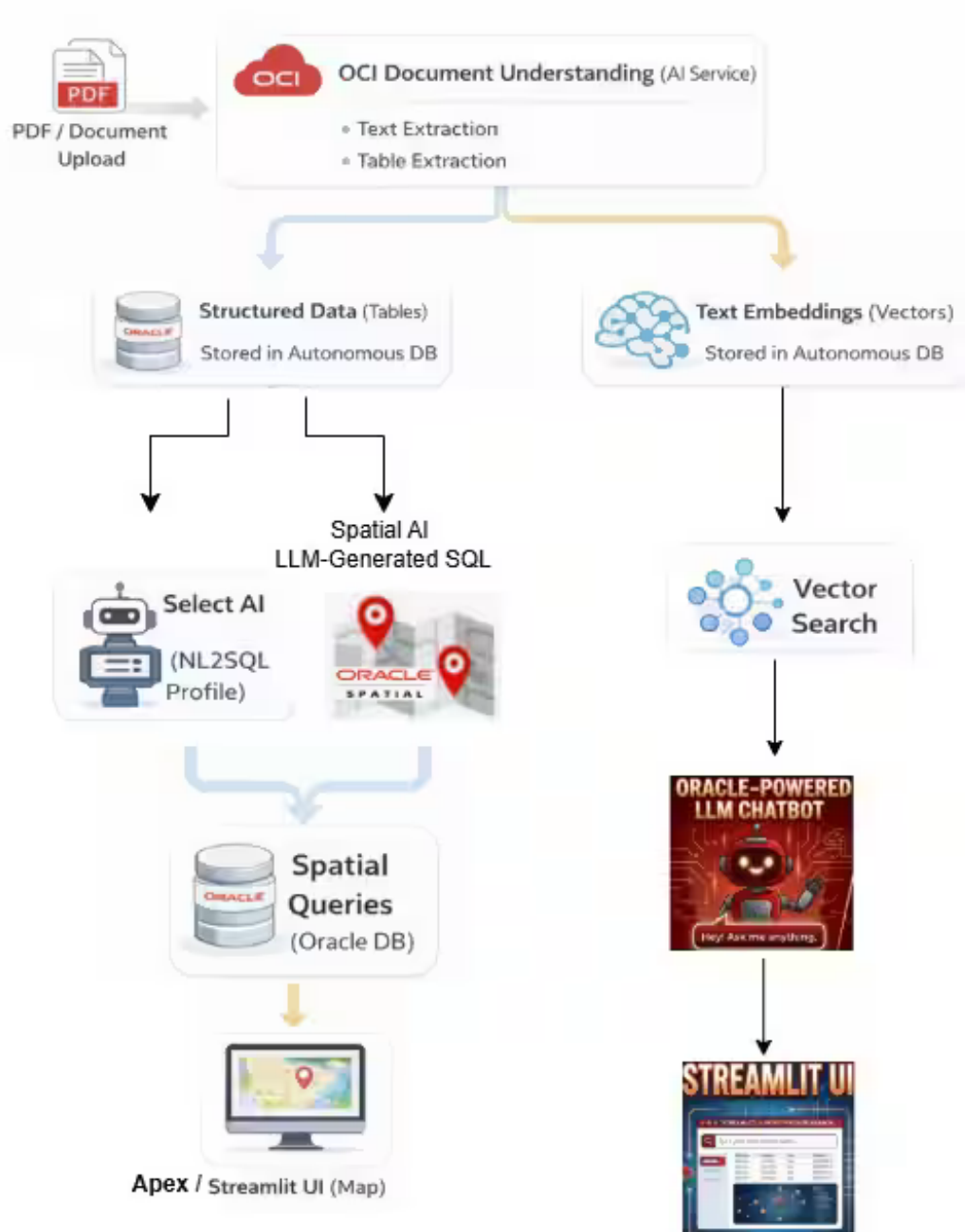
1. Extracts structured data from documents
2. Stores it in Oracle Autonomous Database
3. Uses natural language queries to retrieve insights
4. Performs spatial analysis using LLM-generated SQL
5. Visualizes results on a map

Additionally:

- Text can be vectorized and embedded to perform **similarity searches using natural language**

Architecture Overview

High-level flow of the solution:



Architecture Workflow

1. Upload a sample PDF containing address and geolocation data
2. Extract tabular data and store it in Oracle Autonomous Database
3. Extract text data and store it as embeddings
4. Create Select AI profile for NL2SQL

5. Create another profile for spatial analysis using LLM
5. Execute queries and visualize results using Streamlit

Prerequisites

Create AI Credential

[Copy code snippet](#)

```
BEGIN
DBMS_CLOUD.CREATE_CREDENTIAL(
credential_name => '<cred_name>',
user_ocid => '<your_ocid1.user>',
tenancy_ocid => '<your_ocid1.tenancy>',
private_key => '-----BEGIN RSA PRIVATE KEY-----
...
-----END RSA PRIVATE KEY-----',
fingerprint => '');
END;</your_ocid1.tenancy></your_ocid1.user></cred_name>
```

Ref : <https://docs.oracle.com/en-us/iaas/Content/API/SDKDocs/cliinstall.htm#configfile>

Create Select AI Profiles

[Copy code snippet](#)

```
BEGIN
DBMS_CLOUD_AI.create_profile(
profile_name => 'profile_name',
attributes => '{
    "provider": "oci",
    "credential_name": "cred_name",
    "object_list": [{"owner": "<schema_name>",
        "name": "</schema_name>"
```

Example:

[Copy code snippet](#)

```

BEGIN
DBMS_CLOUD_AI.create_profile(
profile_name => 'PROP_DET_G',
attributes => '{
    "provider": "oci",
    "credential_name": "GENAI_CRED",
    "object_list": [
        {"owner": "APPUSER", "name": "SLA_PROPERTY_DETAILS_G"}]
}');
END;

```

Create Spatial Analysis Profile

Example :

[Copy code snippet](#)

```

BEGIN
DBMS_CLOUD_AI.create_profile(
profile_name => 'SPAT_ANALY_MLLAMA4',
attributes => '{
    "provider": "oci",
    "credential_name": "GENAI_CRED",
    "model": "meta.llama-4-maverick-17b-128e-instruct-fp8",
    "region": "us-chicago-1",
    "oci_apiformat": "GENERIC",
    "temperature": 0,
    "object_list": [
        {"owner": "<schema_name>", "name": "PROPERTY_DETAILS"},
        {"owner": "<schema_name>", "name": "SINGAPORE_AMENTI"}
    ]
}');
END;</schema_name></schema_name>

```

Ref: <https://docs.oracle.com/en-us/iaas/autonomous-database-serverless/doc/dbms-cloud-ai-package.html>

Note: Here we are using the pre-trained model meta.llama-4 , which provides appropriate responses for spatial operators. However, we can use any models based on our requirements. Refer [here](#) for the latest LLM model information.

Python Integration

Config – Initialize the global variables in pycode

[Copy code snippet](#)

Example:

```
CONFIG_PATH = r"C:\Users\Sathishkumar.oci\config"
CONFIG_PROFILE = "DEFAULT"
COMPARTMENT_ID = "ocidl.compartment.oc1"
NAMESPACE = "sehubjapaciaas"
BUCKET_NAME = "DM_bucket"
PREFIX = "aiop"
```

Note: Either you can configure OCI CLI and store all OCI resource information in a config file, or you can assign those values as GLOBAL parameters inside the Python script.

Initialize OCI Clients

[Copy code snippet](#)

```
def init_clients():
    config = oci.config.from_file(CONFIG_PATH, CONFIG_PROFILE)

    ai_client = oci.ai_document.AIServiceDocumentClientCompositeOpenAPI(
        oci.ai_document.AIServiceDocumentClient(config=config)
    )

    object_storage_client = oci.object_storage.ObjectStorageClient(
    return ai_client, object_storage_client
```

Data Processing Workflow

Upload PDF

Extract structured and unstructured data

Store tabular data in database

Store embeddings for semantic queries

The complete python routines can be referred [here](#)

1- Querying with Select AI (NL2SQL)

Set profile:

[Copy code snippet](#)

```
BEGIN
DBMS_CLOUD_AI.SET_PROFILE(profile_name => 'PROP_DET_G');
END;
```

Example: Prompts in natural language

List all records where pincode = 571100

Show properties on street 'Lush on Holland Hill'

Chatbot - OCI Select AI

Ask a question about extracted contact data:

list all records where pincode equal to 571100

Ask Select AI

> Generated SQL

Query Result

	BLOCK_HOUSE	STREET_NAME	STOREY	UNIT_NO	POSTAL_CODE	LAT	
0	100A	BISHAN STREET 13	13	132	571100	1.3	47
1	100B	BISHAN STREET 13	13	133	571100	1.3	48

Ask a question about extracted contact data:

show details of property residing in the street name of 'Lush on Holland Hill'

Ask Select AI

> Generated SQL

Query Result

ID	SNO	FILE_NAME	BLOCK_HOUSE	STREET_NAME	STORE	Select rec to vis:
0	52	10	crsk_address.pdf	32	Lush on Holland Hill	 52

Above, output list all the properties based on prompt using Select AI (NL2SQL)

2 – Spatial Visualization

Convert address into geolocation using Geoapify

Display Points of Interest (PoI)

Visualize using Streamlit map

How it works: Based on the selected record, the corresponding address attributes (such as building name, street name, and pincode) are structured into a JSON payload and sent to the [Geoapify Geocoding API](#). The service resolves the address into precise geographic coordinates (latitude and longitude). Once the location is identified, nearby Points of Interest (Pols) are retrieved and rendered on the map in the application for spatial visualization.

Spatial Visualization

Using Oracle Spatial Geometry

Modify or Enter Address:

100A, BISHAN STREET 13, 132, 571100

Visualize Location

Location Detected: Lat: 1.3461851, Lon: 103.849434769659



Spatial Analysis with Select AI

Switch profile:

[Copy code snippet](#)

```
cursor.execute("""
BEGIN
DBMS_CLOUD_AI.SET_PROFILE(profile_name => 'SPAT_ANALY_MLLAMA4');
END; """)
```

Example: Prompts

Show all amenities near property ID 370 within 3 km

Show hospitals near property ID 52 within 2 km

Generate SQL

Sample Spatial Analytical query Generation using Select AI

[Copy code snippet](#)

```
SELECT DBMS_CLOUD_AI.GENERATE(prompt => 'Show all amenities near prop
profile_name => 'SPAT_ANALY_MLLAMA4',
action => 'showsql') as query
FROM dual;
```

[Copy code snippet](#)

```
SELECT DBMS_CLOUD_AI.GENERATE(prompt => 'Show hospitals near propert
profile_name => 'SPAT_ANALY_MLLAMA4',
action => 'showsql') as query
FROM dual;
```

[Copy code snippet](#)

Call the prompts inside routine in python

```
cursor.execute("""
    SELECT DBMS_CLOUD_AI.GENERATE(
        prompt => :1,
        action => 'runsql'
    ) FROM dual
    """, [system_prompt_spatial])
```

Spatial Analytics query generated by LLM :

Using Oracle Select AI + Oracle Spatial Operators

Ask spatial question:

show all amenity near to property id 370 with in range of 3 km

Run Spatial Analysis

Generated Spatial SQL

```
SELECT
  a."AMENITY_NAME",
  a."LOC".SDO_POINT.Y AS "LAT",
  a."LOC".SDO_POINT.X AS "LON",
  sdo_geom.sdo_distance(a."LOC", p."LOC", 0.005, 'unit=km') AS "DIST_KM"
FROM
  "APPUSER"."SINGAPORE_AMENTIES" a,
  "APPUSER"."SLA_PROPERTY_DETAILS_G" p
WHERE
  p."ID" = 370
  AND SDO_WITHIN_DISTANCE(a."LOC", p."LOC", 'distance=3 unit=km') = 'TRUE'
ORDER BY
  sdo_geom.sdo_distance(a."LOC", p."LOC", 0.005, 'unit=km')
```

Note: Provide system prompts to guide spatial SQL generation using Oracle Spatial operators. You can refer the complete system prompt [here](#)

Spatial Result

	AMENITY_NAME	AMENITY_TYPE	LAT	LON	DIST_KM
7	New Life Coach Doctor	hospital	1.3063	103.8204	2.2463
8	Faculty of Dentistry	school	1.2958	103.7836	2.3693
9	Nursing Training Centre	hospital	1.287	103.802	2.427
10	Urgent Care Centre	hospital	1.2869	103.8016	2.4383
11	National University Hospital	hospital	1.2942	103.7831	2.5257
12	Yong Loo Lin School of Medicine	school	1.296	103.7817	2.5285
13	Pacific Family Clinic	hospital	1.2892	103.8163	2.8077
14	Kurt Hahn House (UWCSEA Dover Campus)	school	1.3026	103.7757	2.8395
15	Maya Angelou House (UWCSEA Dover Campus)	school	1.3034	103.7751	2.8788
16	Nelson Mandela House (UWCSEA Dover Campus)	school	1.3029	103.7752	2.887

Spatial Visualization

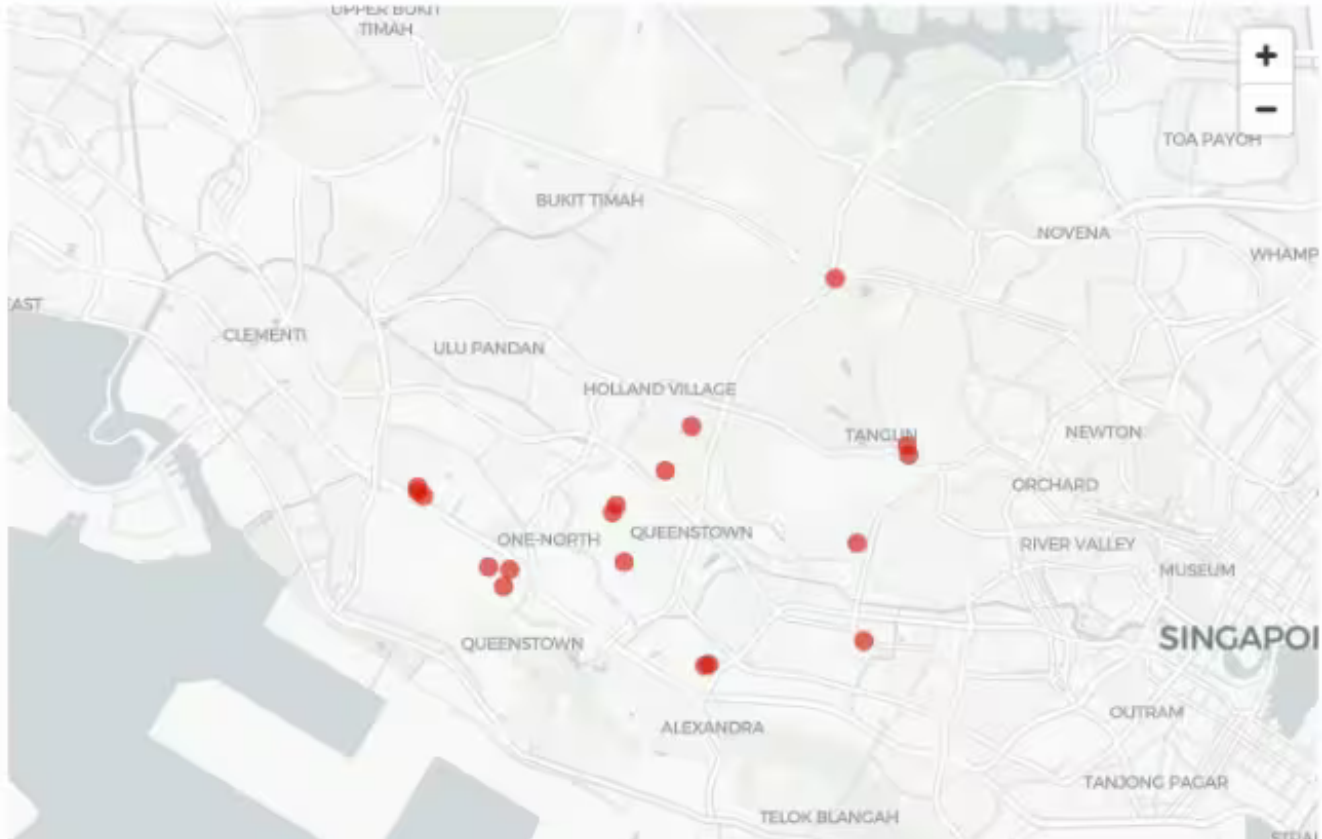
The results (amenities, locations, and distances) are visualised using the following:

Interactive maps

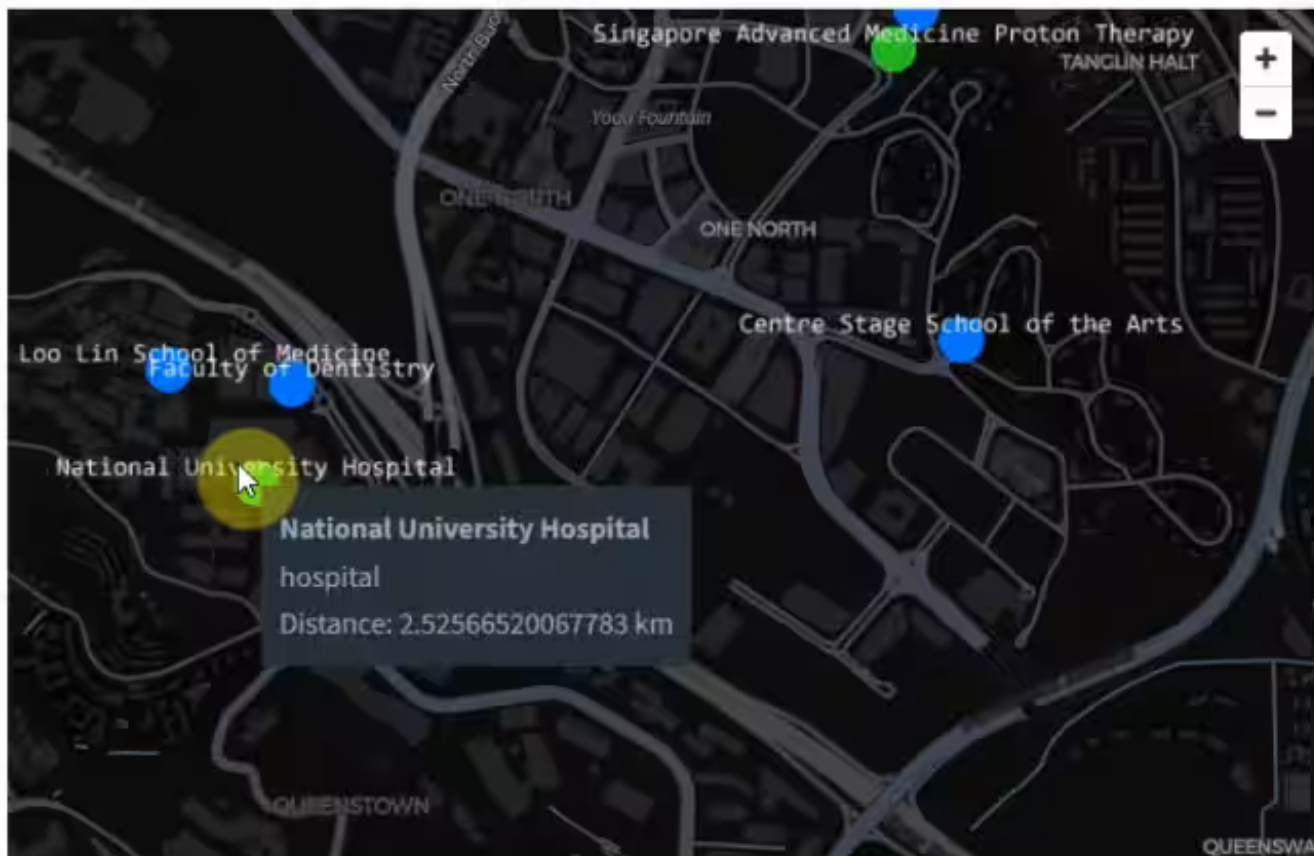
Real-time filtering

Spatial overlays

Spatial Map Result



Spatial Map Visualization



Key Takeaways

- Automate document processing using OCI Document Understanding
- Query structured data using natural language (Select AI)
- Generate spatial queries using LLMs
- Combine relational + vector approaches for better accuracy
- Build end-to-end applications rapidly with Python

Conclusion

By combining **Document AI, Select AI, and Spatial capabilities**, developers can build powerful applications that transform unstructured documents into **location-aware intelligence systems**.

This approach is ideal for:

- Real estate analytics
- Smart cities
- Logistics and supply chain
- Location intelligence platforms

For complete implementation, refer [here](#):



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Sathishkumar is a Master Principal Solution Architect specializing in Oracle Cloud Infrastructure, enabling global customers in their digital transformation journeys. He designs automated, scalable data replication and migration

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